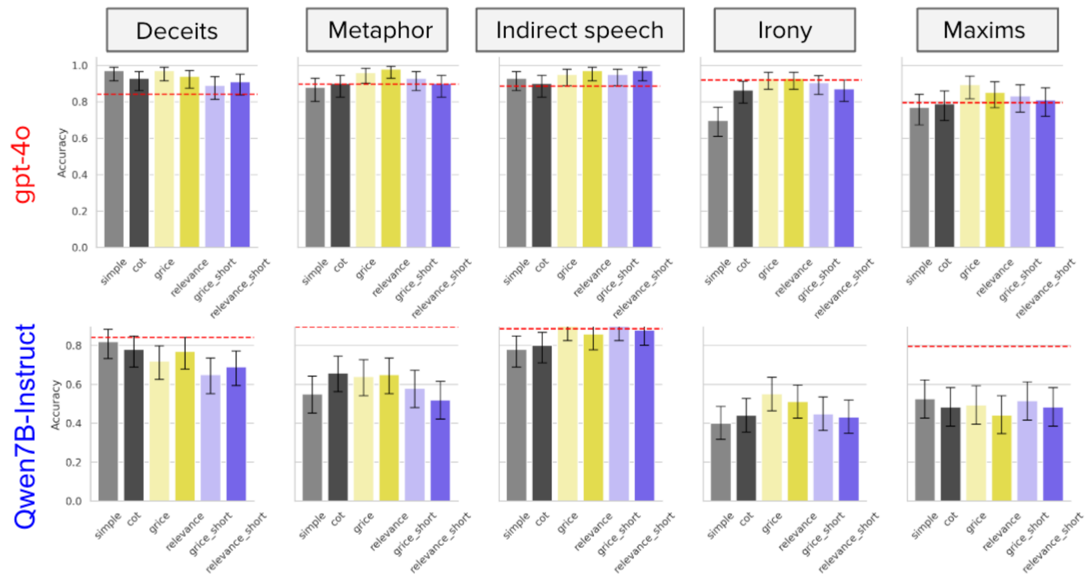

Figure 2: Accuracy of the model for each pragmatic phenomenon included in PRAGMEGA (Hu et al., 2023) when using different methods. Due to space constraints, we present the results for GPT-4o and Qwen2.5-7B-Instruct (for detailed results, including other models, see Appendix §I). The human score is based on (Hu et al., 2023).

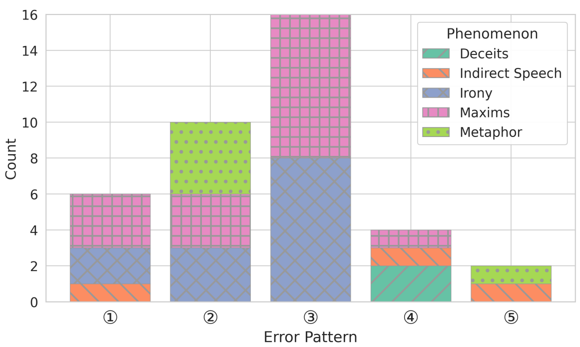

Figure 3: The number of instances for each error pattern by GPT-4o, as described in the main text. A cumulative bar chart represents these counts, including the distribution of each phenomenon within each pattern.

contextual information. If such problems can be considered to lack a "definitive correct answer," then the performance of cutting-edge models like GPT-4o may already be saturated within our experimental setting.

④ Cases where only *Gricean Prompting* was effective This category includes cases where only `grice` and `grice_short` resulted in correct answers, while all other methods failed. The most frequent phenomenon observed in this pattern was *Deceits*. It could be valuable to explore how different results might emerge if theories that focus more on social aspects, such as Politeness Theory (Brown and Levinson, 1987), were incorporated into the prompts.

⑤ Cases where only *Relevance Prompting* was effective This category includes cases where only `relevance` and `relevance_short` resulted in correct answers, while all other methods failed. This pattern was relatively rare, with only two instances equally distributed between the *Metaphor* and *Indirect Speech* phenomena.

5 Additional Experiments: Examination of Confounding Factors

5.1 Motivation and Experimental Settings

In the previous section, we demonstrated that providing an explanation of pragmatic theory can improve the performance of various LLMs on the PRAGMEGA dataset, and furthermore, that this improvement is not merely a result of input or output length effects. We also hypothesized that our prompt might have effectively activated the LLMs' latent knowledge of pragmatic theory. However, the improvement observed with the proposed method may stem from confounding factors rather than the intended pragmatic theories. Specifically, the following concerns remain[4].

1. Even when using a theory that is authoritative in appearance but largely unrelated to the task—rather than a pragmatic theory—the performance might improve to a similar or even

[4]These possibilities were pointed out in comments from anonymous reviewers during the ARR 2025 July Cycle.

Table 1: Actual question examples for some error patterns. **Bold** indicates the correct options. Due to space constraints, examples other than ① and ③ are provided in Appendix §J.

Pattern	Questions	Options
①	Samantha is talking with her dad about her fiance. Samantha notes: "John is an innocent person." Her dad replies: "Undoubtedly, as innocent as a saint." Why has Samantha's dad responded like this?	1. Samantha's dad is impressed with John's innocence. **2. Samantha's dad thinks that Samantha has an incorrect view of her fiance.** 3. Samantha's dad thinks that Samantha's fiance is a saint. 4. Samantha's dad thinks that John is too religious.
③	John is a teacher at an elementary school. When talking with the principal about a new student, who did poorly on her entrance examination, John said, "This one is really sharp." What did John want to convey?	1. The entrance exam is unfair. 2. The pencils need to be sharpened. 3. The student is smart. **4. The student is not very clever.**

greater extent than with the proposed method.

2. Even when using an entirely fictitious "pragmatic theory" that is nonsensical but appears formally plausible, the performance might improve to a similar or even greater extent than with the proposed method.
3. Even without using a pragmatic theory, simply using a general prompt template other than 0-shot CoT might yield performance improvements comparable to or greater than those achieved by the proposed method.

To test these concerns, we ran additional experiments with different prompts for each scenario. Foremost, to investigate the first concern, we conducted an experiment using the following methods, which incorporate a theory unrelated to the task.

X-bar Theory Prompting (`xbar`) In this setting, the prompt instructs the model to "think in line with X-bar theory (**??**)."

Computational Complexity Prompting (`complexity`) In this setting, the prompt instructs the model to "think in line with the theory of computational complexity."

Graph Theory Prompting (`graph`) In this setting, the prompt instructs the model to "think in line with the graph theory."

To test the second concern, we used fictitious pragmatic theories in the following experiments.

Majoritarian Pragmatics Prompting (`majoritarian`) In this setting, the model receives a brief description of a made-up pragmatic theory, Majoritarian Consensus Pragmatics, and is told to reason according to it. This theory claims that an expression's meaning is simply what the majority would interpret it as.

Majoritarian Pragmatics Short Prompting (`majoritarian_short`) In this setting, the prompt instructs the model to "think in line with Majoritarian Consensus Pragmatics," but the description of the theory itself is omitted from the prompt.

Distance Pragmatics Prompting (`distance`) Here, the model is told to reason and answer using a brief description of the fictional theory named Distance Primacy Pragmatics. This theory claims that the primary factor in utterance interpretation is the physical distance (spatial parameter) between speaker and listener, while linguistic content and social context are merely secondary.

Distance Pragmatics Short Prompting (`distance_short`) In this setting, the prompt instructs the model to "think in line with Distance Primacy Pragmatics," but the description of the theory itself is omitted from the prompt.

Additionally, to examine the third concern, we conducted experiments using the following method.

Plan-to-Solve Prompting (`plan`) This setting employs the Plan-to-Solve Prompting method proposed by Wang et al. (2023a). In the prompt, the model is instructed as follows: "Let' s first understand the problem and devise a plan to solve it. Then, let' s carry out the plan and solve the problem step by step."

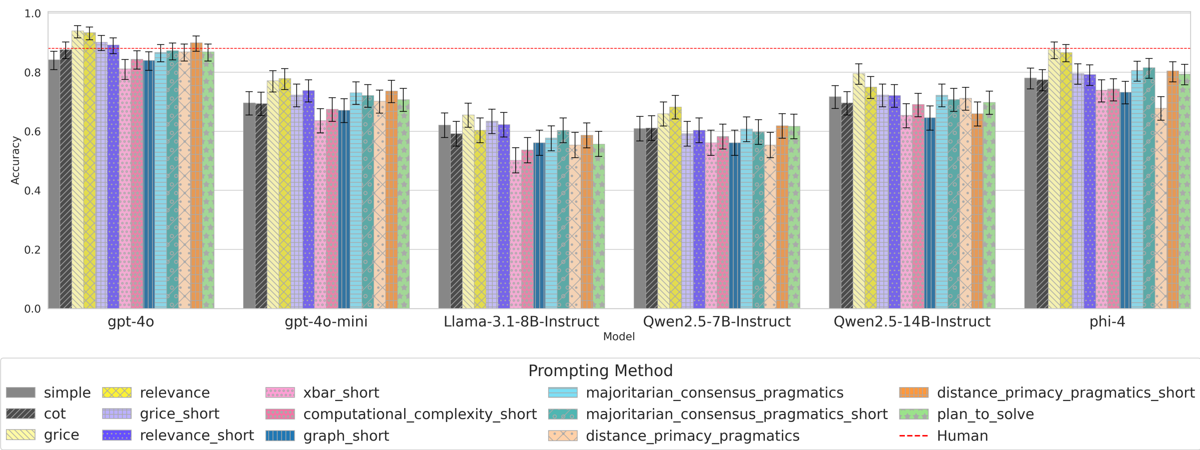

Figure 4: Results of the additional experiments

The specific prompts used for each method are presented in Tables 4, 5, and 6 in Appendix D. The models and detailed experimental settings are identical to those described in the previous section.

5.2 Results and Analysis

The results of the additional experiments are shown in Figure 4. Detailed results are presented in Table 8 in the appendix. Among all combinations of models and additional prompting methods, none achieved a higher accuracy than the proposed methods, `grice` and `relevance`. In other words, no experimental evidence was found to support the concerns above.

However, among the additional methods, some achieved higher accuracy than `grice_short` and `relevance_short`. In particular, `distance_short` frequently exhibited such cases. This may suggest that even if a theory is entirely fictitious in name, the model can sometimes find useful meaning or interpretation in the prompt and solve the task, as long as the prompt is not absurd or incoherent.

Providing prompts based on well-known but task-irrelevant theories—whose content the model is likely to have encountered during pretraining—tended instead to decrease the model' s performance on our task. Furthermore, using a general prompt template, such as `plan`, sometimes resulted in either decreased or improved performance compared to the baseline methods.

6 Conclusion

In this paper, we proposed an instance-agnostic method for pragmatic reasoning tasks by incorporating an overview of linguistic pragmatic theories into the model's prompt. Experimental results demonstrated that the proposed methods improve performance on pragmatic reasoning tasks without providing instance-dependent information to the model in a top-down manner.

As a direction for future work, it is desirable to develop pragmatic reasoning tasks that require consideration of richer and more extended contexts, followed by experiments using such tasks. A limitation in PRAGMEGA and other previous studies is that the instances they handle often lack sufficient complexity compared to real-world pragmatic phenomena, or the contextual information relevant to interpretation is insufficiently rich. To address these issues, creating datasets that incorporate more complex and contextually rich pragmatic phenomena, along with tasks utilizing these datasets, is necessary for a more precise analysis of model capabilities and limitations. The dataset used in (Shisen et al., 2024), which is based on a Chinese sitcom, represents a promising approach to addressing these challenges. Focusing on this direction and developing more comprehensive datasets and benchmarks will contribute significantly to the further advancement of this research field.

Finally, exploring the approach of applying discussions from specific domains as prompt templates, as we have done in other domains and tasks, would contribute to optimizing model performance strategies and exploring the nature of the tasks themselves in those use cases.

Limitations

One limitation of our research is that we have not been able to verify whether the proposed method is effective when applied to more upstream tasks or